

Neurovascular Disease in Women

How sex, hormones and pregnancy reshape the cerebral circulation

A visual summary



A circulation under hormonal control

The vessels of the brain do not behave the same way in every body.

Sex hormones, the clotting shifts of pregnancy and sex-specific risk factors all tilt the balance between flow, clotting and bleeding.

Pregnancy & postpartum

A pro-thrombotic window

Hormones & contraception

Estrogen tunes vascular risk

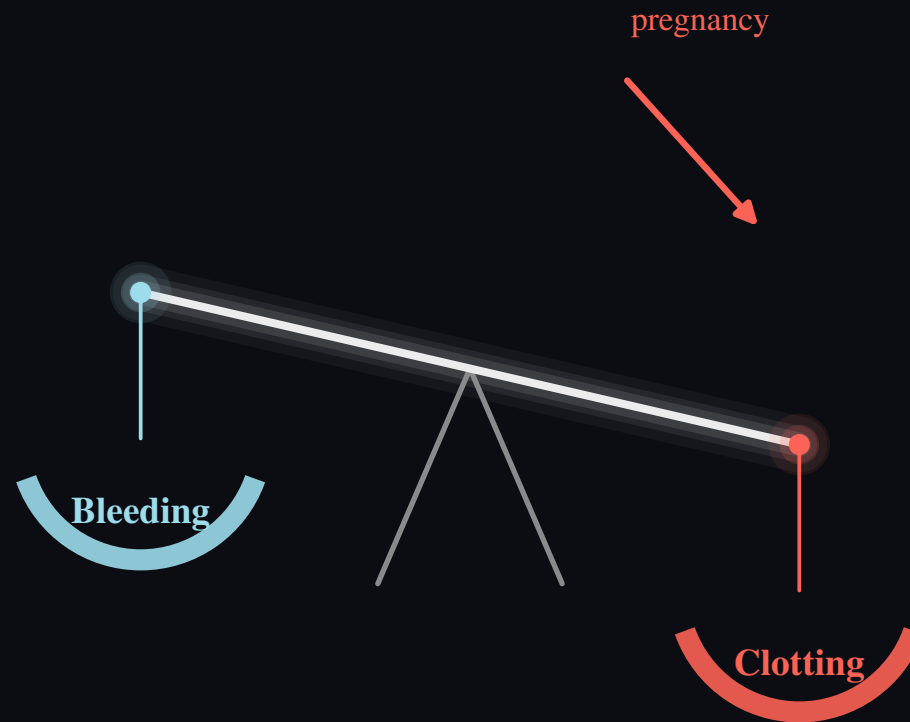
Lifelong risk

Pregnancy as a cardiovascular stress test

Pregnancy tilts hemostasis toward clotting

Clotting is dialled up to limit blood loss at delivery — an adaptive shift.

The same shift makes the cerebral vessels prone to thrombosis.



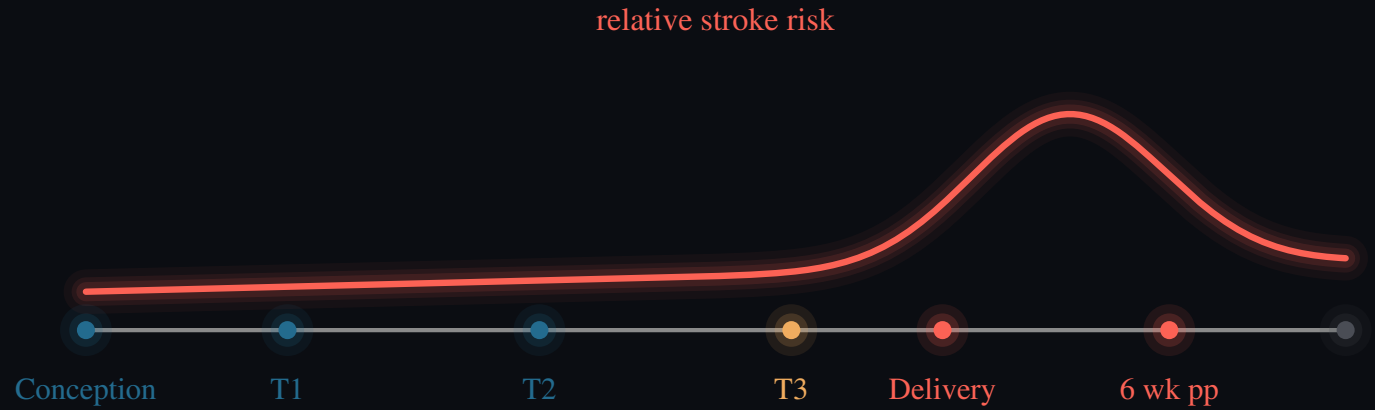
- Hypercoagulable state
- Volume shifts & dehydration
- Venous stasis

Highest risk: 3rd trimester
→ first 6 weeks postpartum

Stroke in pregnancy: rare, but it can be devastating

16–41

ischemic strokes
per 100,000 pregnancies

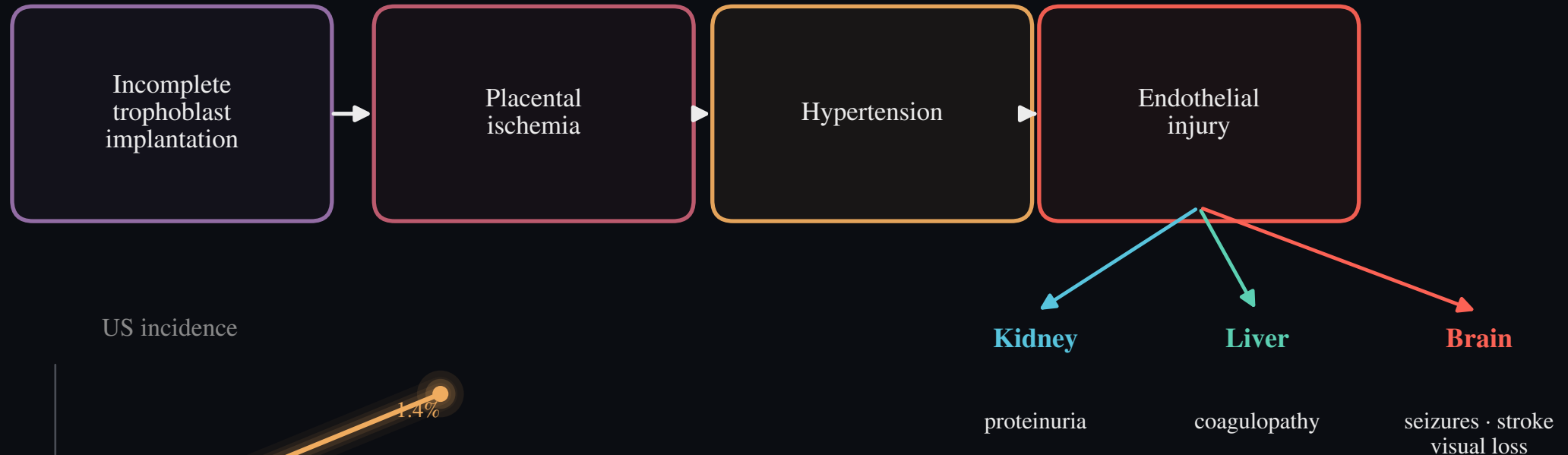


The hypercoagulable state peaks late — the late third trimester and the early postpartum weeks carry the greatest cerebrovascular risk.

Pre-eclampsia / eclampsia: a vascular disease

New-onset hypertension + end-organ injury after 20 weeks.

Eclampsia = pre-eclampsia with new seizures.



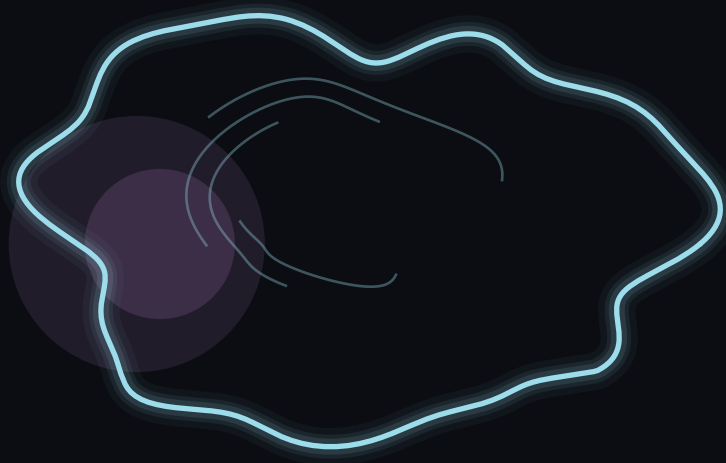
US incidence



PRES: the posterior brain loses its grip

The posterior circulation autoregulates poorly against surging pressure.

Fluid leaks into posterior white matter — posterior reversible encephalopathy syndrome.



posterior
white-matter edema

- Headache
- Visual changes
- Seizures
- Confusion

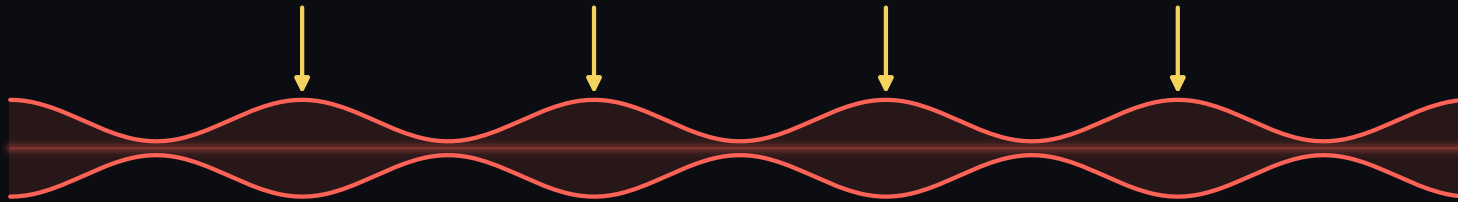
Usually reversible with prompt
blood-pressure control.

RCVS: arteries clamp down in a thunderclap

A sudden, maximal-at-onset headache with segmental arterial narrowing.

Monophasic — it constricts, then resolves over weeks.

segmental narrowing ('string of beads')



thunderclap
headache

Triggers:

vasoactive drugs · SSRIs/SNRIs · sympathomimetics · triptans · ergots · cannabis · cocaine

Strong overlap with pre-eclampsia / eclampsia.

Cerebral venous thrombosis: the drainage clots

Pregnancy's hypercoagulability can clot the brain's venous sinuses.



- Peak window: late pregnancy
through the first 6 weeks postpartum
- Headache · seizures · venous strokes
- Confirm with MR / CT venography
- Treat with anticoagulation (LMWH in pregnancy)

Migraine with aura + estrogen = stroke risk

Migraine is three times more common in women, and aura carries its own small, independent risk of ischemic stroke.

Migraine with aura = absolute contraindication to combined hormonal contraception

Combined hormonal contraception raises stroke risk with the estrogen dose — and that risk is multiplied by aura.



baseline

×



migraine
with aura

×



+ estrogen
contraception

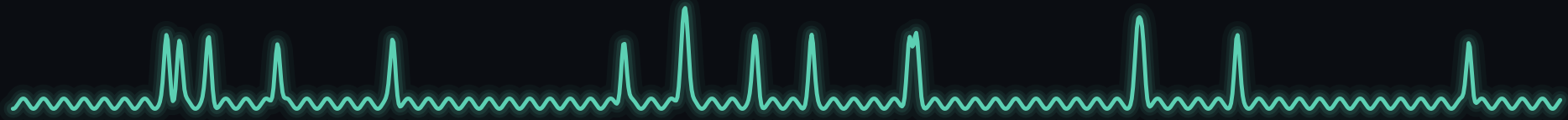
Pregnancy is a cardiovascular stress test

Complications unmask the women whose vessels are already vulnerable —
a decades-long window to intervene before stroke ever happens.



A pregnancy history is part of every woman's stroke-risk assessment.

Atrial fibrillation hits women harder



annual stroke incidence
in nonvalvular AF

6.2%



Women

4.2%



Men

- Female sex adds a point to the stroke risk score
- Worse deficits and poorer quality of life

Lupus & antiphospholipid antibodies

Autoimmune disease is overwhelmingly a woman's disease — and lupus reaches the brain through its blood vessels.

78%

of autoimmune
disease is in women

Lupus: 7 ×
more common
in women

**Antiphospholipid
syndrome**

hypercoagulable state

**Libman–Sacks
endocarditis**

cardioembolic stroke

**Hypertension &
coagulopathy**

small-vessel disease

→ **stroke**

What to carry away



Pregnancy is pro-thrombotic.

The late third trimester and early postpartum carry the highest stroke risk.



Pre-eclampsia is vascular.

Endothelial injury drives PRES, seizures, hemorrhage and stroke.



Think of the syndromes together.

PRES, RCVS and CVT overlap in the peripartum brain.



Estrogen tunes risk.

Migraine with aura contraindicates combined hormonal contraception.



Pregnancy predicts the future.

Its complications forecast stroke decades later — a window to act.

Reference

Primary source

O'Neal MA. A Review of Women's Neurology.

The American Journal of Medicine. 2018;131(7):735–744.

<https://doi.org/10.1016/j.amjmed.2017.11.053>

Figures are original schematic illustrations created for this summary.

